Molecular Biology Reports (2023) 50:3023–3033
https://doi.org/10.1007/s11033-022-08171-w

ORIGINAL ARTICLE

Potentiation of Folate-Functionalized PLGA-PEG nanoparticles loaded with metformin for the treatment of breast Cancer: possible clinical application

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Received: 16 October 2022 / Accepted: 1 December 2022 / Published online: 20 January 2023
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Abstract

Aim: Folate receptor expression increase up to 30% in breast cancer cells and could be used as a possible ligand to couple to folate-functionalized nanoparticles. Metformin (Met) is an anti-hyperglycemic agent whose anti-cancer properties have been formerly reported. Consequently, in the current study, we aimed to synthesize and characterize folate-functionalized PLGA-PEG NPs loaded with Met and evaluate the anti-cancer effect against the MDA-MB-231 human breast cancer cell line. **Methods:** FA-PLGA-PEG NPs were synthesized by employing the W1/O/W2 technique and their physicochemical features were evaluated by FE-SEM, TEM, FTIR, and DLS methods. The cytotoxic effects of free and Nano-encapsulated drugs were analyzed by the MTT technique. Furthermore, RT-PCR technique was employed to assess the expression levels of apoptotic and anti-apoptotic genes. **Result:** MTT result indicated Met-loaded FA-PLGA-PEG NPs exhibited cytotoxic effects in a dose-dependently manner and had more cytotoxic effects relative to other groups. The remarkable down-regulation (hTERT and Bcl-2) and up-regulation (Caspase7, Caspase3, Bax, and p53) gene expression were shown in treated MDA-MB-231 cells with Met-loaded FA-PLGA-PEG NPs. **Conclusion:** Folate-Functionalized PLGA-PEG Nanoparticles are suggested as an appropriate approach to elevate the anticancer properties of Met for improving the treatment effectiveness of breast cancer cells.

Keywords Metformin · Folate Functionalized-PLGA-PEG · Nano-carrier · Breast cancer

Introduction

Breast cancer the fifth cause of cancer deaths in the world has exceeded lung cancer, with an estimated 2.3 million cases and 685,000 deaths in 2020, and the cases are expected to reach 4.4 million in 2070 [1–3]. Age, race, early/late menopause, and alcohol consumption are considered the main risk factors for breast cancer [4].

Metformin (Met) is one of the safest and most effective anti-hyperglycaemic agents currently employed as first-line treatment for type 2 diabetes and is effective as monotherapy and in combination with other glucose-lowering drugs [5–7]. Different applications of Met were reported over time. Anti-cancer roles of Met have been studied for many years, new evidence suggests that Met may inhibit malignancies and metastases as well as arrest the cell cycle on the G1 checkpoint and change the receptor tyrosine kinase (RTK) signaling pathway [8–11]. Cell cycle interference

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and signaling pathways such as the PI3K pathway is a therapeutic targets in breast cancer like other malignancies [12].

Several mechanisms are described for the anti-cancer effects of Met which are associated with AMPK and mTOR signaling pathways that lead to the inhibition of cell growth and proliferation in breast, endometrial, lung, liver, gastric, and thyroid cancer cell lines. Studies have revealed that survival rates in diabetic breast cancer patients who used Met are remarkably higher relative to the others [13–15]. One of the most developed nano-polymers for drug delivery over the years is PLGA (poly lactic-co-glycolic acid). Important properties of this polymer include bio-compatibility, biodegradability, EMA (European Medicines Agency) and FDA (U.S. Food and Drug Administration) approved, well-described production process, the possibility of modification for better interactions, and protection drug from degradation [16, 17]. Glycolic acid and lactic acid, the components of PLGA, are easily metabolized in the Krebs cycle and from the toxicological point of view are safe [18]. In order to deliver a sufficient concentration of systemically administered drugs to the target tissues, they must circulate in the bloodstream as long as possible [19]. Nanoparticle PEGylation has also proven to be a very effective approach to improving the systemic delivery of therapeutic agents. In addition, the use of PEGylation nanoparticles overcomes various extracellular barriers associated with other methods of administration, from mucosal delivery to tissue delivery [20–22]. PEG (polyethylene glycol) is a non-toxic and water-soluble compound that can covalently bind to proteins [23]. PEG changes the properties of the materials to which it binds. Conjugation of PLGA nanoparticles (NPs) with polyethylene glycol (PEG) improves the pharmacological properties [14].

Chemotherapy is unable to differentiate between cancer and normal cells, the combination of nanoparticles and chemotherapy for drug delivery is a new therapeutic approach that can solve the problem of traditional methods including poor water solubility, lack of targeting, toxicity problems, and nonspecific distribution [24, 25]. Nowadays drug selectivity for cancer cells is a big challenge. Previously, antibodies were used to increase selectivity, but now studies show that folic acid can be used for this purpose [26]. Folate receptor expression increase up to 30% in breast cancer and is used as possible ligand to couple to nanocarriers [26, 27].

Folic acid is an attractive target for the selective targeting of folate receptor alpha (FRα) expressing cancer cells. FRα is a membrane-bound protein richly expressed on the surface of many human tumor cells such as breast cancer cells, while its expression is limited in normal cells [28].

MDA-MB-231 breast cancer cells are enhanced with the Folate receptor expression on their membrane surface. Therefore, the overexpression of FRα on the surface of MDA-MB-231 could be thought an effective target for the targeted delivery of anti-cancer agents [29].

Recent studies have shown that overexpression of hTERT or the presence of hTERT mutations can be related to cancer cell migration, invasion, and metastasis. Also, it has been demonstrated that in 90% of breast carcinomas and in 85% of human cancers, telomerase is activated. Thus, suppression of the expression of the hTERT gene and telomerase activation can be a promising approach in cancer therapy [10, 15, 30–33].

BAX / BAK are pro-apoptotic molecules that cause cell death during the apoptotic process. The BCL-2 family of proteins is involved in the suppression of BAX / BAK, and overexpression of BCL-2 is associated with a variety of malignancies, making BCL-2 a promising anti-cancer treatment. Caspases are a type of cysteine protease that plays an important role in apoptosis and is activated in response to cell death signals. Caspases 3 and 7 are the main caspases of execution among caspases [34, 35].

For this purpose, in this study, we aimed to develop the Folate-Functionalized PLGA-PEG nanoparticles loaded with metformin to evaluate the cytotoxicity and expression level of the p53, Bcl-2, Bax, Caspase7, Caspase3, and hTERT genes in the MDA-MB-231 breast cancer cell line.

Materials and methods

Chemicals and reagents

MDA-MB-231 cell lines (Human breast cancer) were obtained from the Pasteur Institute of Iran and stored according to the recommendations. Metformin (1,1-dimethyl biguanide hydrochloride), Folate(FA), polyvinyl alcohol (PVA), DL-lactide (LA), dichloromethane (DCM), Dimethyl sulphoxide (DMSO), glycolide (GA), polyethylene glycol (PEG, MW 4000), and MTT powder (3(4, 5-dimethyl thiazol-2-yl) 2, 5-diphenyl-tetrazolium bromide) were purchase from Sigma-Aldrich (St Louis, USA). First Strand cDNA synthesis and Real-time PCR (RT-PCR) Kit were provided from Takara (Shiga, Japan). Dynamic light scattering (DLS), Fourier transform infrared (FTIR) spectroscopy, TEM, and field emission scanning electron microscopy (FE-SEM) techniques were employed to qualify and characterization of the produced nano-carriers.

Preparation of FA- PLGA PEG- NPs

Ring-opening polymerization method under a nitrogen atmosphere and in presence of stannous octoate [Sn (Oct) 2: stannous 2-ethylhexanoate] as a catalyst was applied for the preparation of PLGA PEG NPs, Briefly, DL-lactide (2 g),

and glycolide (0.6 g) were heated to 140 °C in a bottleneck flask which connected to a vacuum system. Then, 0.05% (w/w Sn (Oct) 2 was added and the temperature of the reaction mixture was increased to 180 °C and kept for 4 h.

PLGA (2 g) and methylene chloride solution were activated using DCC (160 mg) and NHS (115 mg), which were performed at ambient temperature and under a nitrogen atmosphere for 24 h. Then, the prepared solution was passed through a filter and precipitated by pouring into ice-cold diethyl ether and then the activated PLGA with free terminal carboxyl groups was entirely dehydrated under vacuum. Activated PLGA (1 g) was liquefied through methylene chloride (8 ml) and then lightly put into PEG-bis-amine (100 mg) liquefied in methylene chloride (2 ml), which was stirred slowly. The reaction time was kept for 6 h in a nitrogen atmosphere. Frozen cold ethyl ether was added and the resulting solution precipitated. Amine-terminated di-block copolymer, PLGA–PEG– NH2, was a sedimentary product that was passed through a filter and dehydrated. Synthesis of FA- PLGA PEG- NP was performed through binding PLGA– PEG–NH2 di-block copolymer by an actuated folic acid.

For this reason, concisely, di-block copolymer (500 mg) was dissolved in DMSO (5 ml) then DCC (13 mg) and folic acid (13 mg) were added and mixed.

The reaction time was 7 h at 25–28 degrees and next mingled with cold methanol (100 ml) and filtered with a paper filter. The resulting precipitate on the filter was collected and desiccated under vacuum, next was dissolved in DCM (50 ml). After centrifugation at 21,000 g, conjugated FL was dissolved. The final supernatant was dried under a vacuum.

Folate-decorated PLGA-PEG loaded with metformin

The double emulsion (w/o/w) procedure was utilized to manufacture PLGA-PEG nanoparticles loaded with Met. In this synthesis mechanism, first, the PLGA (-COOH) and PEG (-OH) functional groups were linked by removing a water molecule, then the organic folic acid compound was added to PLGA-PEG. The FA (-COOH) and PLGA-PEG (-OH) functional groups were linked by removing a water molecule. According to this process, by removing two water molecules and adding an organic compound, the mechanism proceeded in W/O/W form. Briefly, FA- PLGA PEG- NP (200 mg) was dissolved in dichloromethane in a volume of 5 ml. then Met) 20 mg (, which previously was solved in 20 ml of polyvinyl alcohol (PVA, 1% w/v), poured to the organic part of the tube and by employing ultrasonic, it was emulsified (Sonoplus, HD 2070; Bandelin Electronics, Berlin, Germany) for 55 min at 55% strength. By means of a rotary evaporator (Heidolph Instruments, Hei-VAP series), DCM was evaporated under vacuum at 40 degrees. Finally, the Met-NPs were collected with centrifugation at 15,000 rpm for 40 min.

Physicochemical characterization of NPs

Analysis of particle size, distribution, and zeta potential

Via the DLS technique, the hydrodynamic average size (nm) and surface charge (ζ-potential, mv) of obtained NPs were assessed. For this reason, after suspension of NPs in distilled water, the NPs were sonicated for 5 min, and then Zetasizer Nano ZS (Malvern Instruments Ltd., Malvern, UK) was employed to determine the size and zeta potential which had a helium-neon laser ray at a constant dispersion corner of 90 and a wavelength of 633 nm.

NPs morphology

Transmission electron microscopy (TEM) and field emission scanning electron microscopy (FE-SEM) LEO 1430 VP were employed to evaluate the shape and surface morphology of NPs.

FTIR analysis

Fourier Transformed Infra-Red Spectroscopy (FTIR) information of drugs in free form and loaded in nano-particles that constructed in KBr tablets were documented on a Bruker Tensor 270 spectrometer (from 4000 to 400 cm − 1).

Evaluation of drug encapsulation efficiency

After synthesis, the supernatant of Nano-formulations was isolated and filtered by applying Amicon centrifugal tube (molecular weight cut-off 3000 Da) to evaluate the Met loading efficiency of the PLGA-PEG NPs and FA-PLGA-PEG NPs. The amount of non-entrapped Met was evaluated by high-performance liquid chromatography (HPLC) at maximum wavelengths of Met at 237 (mobile phase composed of methanol and 0.02 M phosphoric acid (50: 50, v/v) with a flow rate of 1 ml/min, and the temperature of 30 °C) and via calibration equations, the quantity of drug in the filtrate was determined. The encapsulated drug is represented by two methods of drug loading (DL) and trapping efficiency (EE), which are shown with Eqs. (1) and (2), respectively.

$$\mathrm{DL} = \frac{\text{Weight of drug in NPs}}{\text{total NPs weight}} \times 100\% \qquad \text{(Eq. 1)}$$

$$EE = \frac{\text{Weight of drug in NPs}}{\text{total Weight drug}} \times 100\% \quad \text{(Eq. 2)}$$

In vitro drug release

Drug release pattern from FA- PLGA PEG- NPs was disclosed with dialysis assay. In summary, 25 mg of Met-loaded FA- PLGA- PEG- NPs was distributed in 5 mL PBS and transported to a dialysis membrane tube with MW cut off 3000 and located in PBS in a volume of 30 ml which was incubated by a stirrer in 120 rpm incubated at 37 °. For 180 h, 3 ml of PBS in the several time periods was exchanged with an identical amount of fresh PBS, and the maximum absorption wavelength of 237 nm for Met was used to measure the concentration of drugs released in the previous medium.

Cell culture and morphological changes

MDA-MB-23 cell lines were purchased from the Pasteur Institute cell culture collection (Tehran, Iran). MDA-MB-23 cells were cultivated in RPMI 1640 medium complemented with 10% of FBS, penicillin G (0.05 mg/mL), and streptomycin (0.08 mg/mL) and then incubated in the sterile flasks at 37 ºC with humidified atmosphere under 5% CO2. To study the morphological change, MDA-MB-231 cancer cells were seeded (5 × 105 cells) in a 6-well plate and incubated for 24 h. Afterward, treated with free Met, Met-loaded PLGA-PEG NPs, Met-loaded FA-PLGA-PEG NPs, and their corresponding blank NPs for 48 h, and the cellular surface morphology changes were captured via the camera-equipped microscope.

In vitro quantitative cellular uptake of NPs

To measure the intracellular drug content quantitatively, MDA-MB-231 cells were seeded at a density of 5 × 105 per well in 6-well plates. After 24 h incubation, the medium was replaced with a fresh medium, and drug-loaded NPs were added and incubated at 37 °C for 4 h. After incubation, the medium was removed and the cells were washed twice with PBS to remove unauthorized NPs. Finally, Cells were lysed for 5 min in a water bath (PBS and 1x triton-100 solution) using an ultrasonic lyser. The concentration of metformin was assessed by the HPLC technique.

Cytotoxicity assay

To evaluate the cytotoxic efficacies of Met in free form and loaded in FA-PLGA-PEG NPs, MTT test was used. MTT assay is based on the ability of mitochondrial succinate dehydrogenase enzyme in survival cells to convert yellow water-soluble tetrazolium salt into insolvable purple-colored formazan compound. In summary, 104 MDA-MB-23 cells were transported to 96-well plates and put into the incubator for 24 h at 37 ◦C in a humid atmosphere including 5% CO2. After 24 h, several concentrations of Met in free, loaded in FA-PLGA-PEG NPs, and PLGA-PEG NPs form were used to treat cells, and then cells were put in an incubator for 48 h subsequently. Following 48 h disclosure time, the cell culture medium was completely evacuated and then replaced with 200 µl of MTT dilution and then the plates were sealed with aluminum foil and placed in an incubator for 4 h at 37 ° C. After that, the previous step, all substance of the wells was removed and 200 µl pure DMSO was poured then plates were incubated for 20 min. In the end, cell viability was assessed by measuring the absorption of each well by an ELISA plate reader (Bio-Tek Instruments) at 570 and 630 nm as reference wavelengths.

Real-time PCR

In this study, RT-PCR technique was used to evaluate the expression mRNA level of Bcl-2, Bax, Caspase7, Caspase3, p53, and hTERT genes. Various concentrations of drugs in free form and drug-loaded NPs were used to treat MDA-MB-23 cells.

After 48 h incubation and drug disposal time, the cells were gathered and all amount of RNA was extricated based on the manufacturer's instructions by using of Trizol reagent. Subsequently, Namedrop ND1000 spectrophotometer (OD260/280 ratio) was used to evaluate the quality of isolated RNA. In addition, the quantity of extracted RNA was assessed by applying agarose gel electrophoresis containing 1.5% agarose. Based on the producer's protocol, the first-strand cDNA synthesis kit (Fermentas, Vilnius, Lithuania) was used to synthesize complementary DNA (cDNA) from RNA. Then RT-PCR technique and specific primers based on the manufacturer's instructions were used to measure the expression level of the studied genes.

RT-PCR reaction was accomplished in this way: First, primary denaturation was performed at 95° C for 10 min, then denaturation stage was performed at 95 ° C for 15 s, annealing stage for 30 s at 60 ° C, expansion for 30 s at 72 ° C, and melting stage was performed at 95–95 ° C. At last, the housekeeping gene (GAPDH) was used to normalize the expression level of studied genes, and calculations were performed through the 2-ΔΔCT formula.

Statistical study

Graph Pad Prism statistical software was used for statistical analysis of the obtained results. All obtained data are

Table 1 Characterization of PLGA-PEG, Met-loaded PLGA-PEG and, Met loaded FA-PLGA-PEG NPs. The results are presented as the average of three independent measurements plus the standard deviation (n = 3)

Groups	Particle size (nm)*	Polydispersity Index	Zeta potential (mV)*
PLGA-PEG NPs	176 ± 3.71	0.147	−29.6 ± 2.1
Met-PLGA-PEG NPs	242 ± 6.28	0.133	−32.1 ± 4.6
Met-FA-PLGA-PEG NPs	**269 ± 4.42**	**0.098**	**−33.6 ± 3.5**

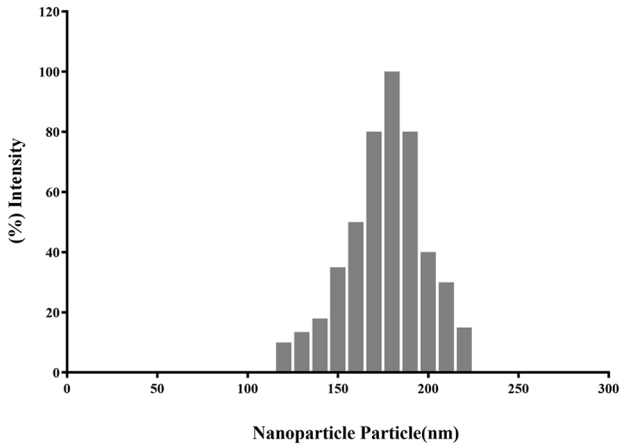

Fig. 1 Characterization of Met-loaded FA-PLGA-PEG NPs size distribution using Dynamic light scattering (DLS) Met-FA-NPs showed a larger diameter compared to drug-free NPs and Met-NPs.

revealed as mean ± SD from at the minimum three independent experiments. A statistical study was done between groups using the ANOVA test and statistical significance was described when the p-value was less than 0.05.

Results and discussion

Characterization of drug-loaded FA- PLGA PEG- NPs

Size, morphology and drug loading

Metformin was loaded in FA- PLGA-PEG through Oil/Water process, which is a common technique for producing therapeutic molecules-loaded polymeric NPs [36]. NPs with suitable physicochemical and biological properties can be easily absorbed by cells and used in drug delivery. NPs should be harmless and biocompatible [37]. Among NPs, PLGA-PEG is commonly used for the delivery of several drugs due to its biodegradability and biocompatibility characteristics [38]. Since kidneys can quickly remove NPs smaller than 10 nm by filtration and the reticuloendothelial system (RES) can easily detect cases larger than 300 nm, this type of NPs can be removed from the bloodstream. Therefore, the produced

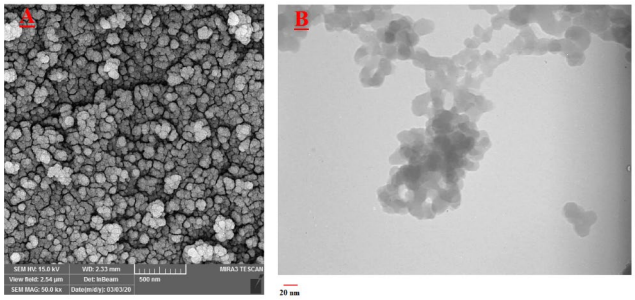

Fig. 2 Characterization of Met loaded FA-PLGA-PEG NPs size and morphology using Field emission scanning electron microscopy (A), and Transmission electron microscopy (B)

NPs must be in the right size and negative surface charge [39]. DLS technique was employed to assess the size of particles and dissemination of NPs. As shown in the Table 1, the mean size of drug-free NPs, Met-NPs, and Met-FA-NPs were 176 ± 3.71, 242 ± 6.28, and 269 ± 4.42 respectively. Compared to Met-FA-NPs and Met-NPs, drug-free NPs showed smaller diameters and larger poly dispersion (Fig. 1). The morphology of drug-loaded NPs was examined with TEM and SEM techniques. The images of Met-NPs and Met-FA-NPs confirmed the spherical shape and uniform distribution of the drug and indicated an average diameter of about 210 nm (Fig. 2 A and Fig. 2B). The drug encapsulation efficiency and loading capacity range (EE) for Met-FA-NPs and Met-NPs were about 83.46%, 13.7 ± 2.9, and 64.31%, 7.3 ± 2.5, respectively. The formation of Met-loaded FA-PLGA PEG-NPs was demonstrated by means of FTIR spectroscopy. The FTIR spectrum of free Met demonstrated two usual peaks at 3146 cm^{-1} and 3290 cm^{-1} assigned to the N–H primary stretching vibrational mode and indicated a peak at 3336 cm^{-1} due to the N–H secondary stretching, and specific peaks at 1626 cm-1 and 1567 cm-1, attributed to C-N stretching. The visible spectrum of Met-loaded FA-PLGA PEG- NPs demonstrated usual bands at 3146, 3290, 2250, 1589.8 cm^{-1} which are allocated to N-H asymmetric stretching, N-H symmetric stretching, C = N stretching, and N-H bending groups (Fig. 3), Furthermore, formation of an amide bond between the amino group and the carboxyl group of folic and PLGA–PEG-NH2, resulted in the slightly salient peak at 1623 cm^{-1}. The results display that Met has been successfully combined with FA-PLGA PEG-NP.

In vitro quantitative cellular uptake of NPs

According to the results, cellular uptake of Metformin from FA-PLGA-PEG NPs was increased compared to PLGA-PEG NPs, which is probably due to the long-term interaction of coated folate with the cell membrane protein of FRα, and facilitates the maximum release of drug-carrier complex

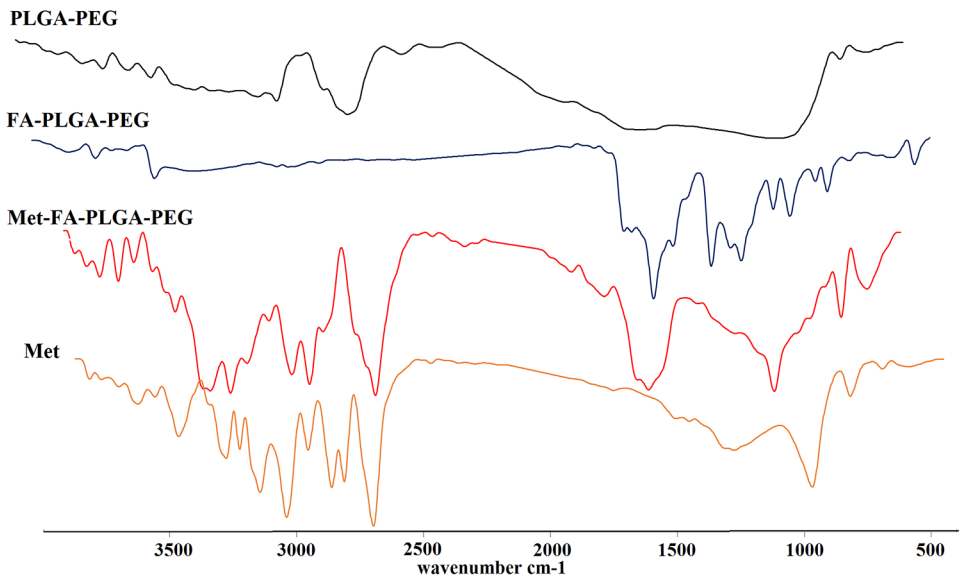

Fig. 3 IR spectra of Met, Met-loaded PLGA-PEG NPs, and Met loaded FA-PLGA-PEG NPs. The results display that Met has been successfully combined with FA-PLGA PEG-NP and PLGA-PEG NPs.

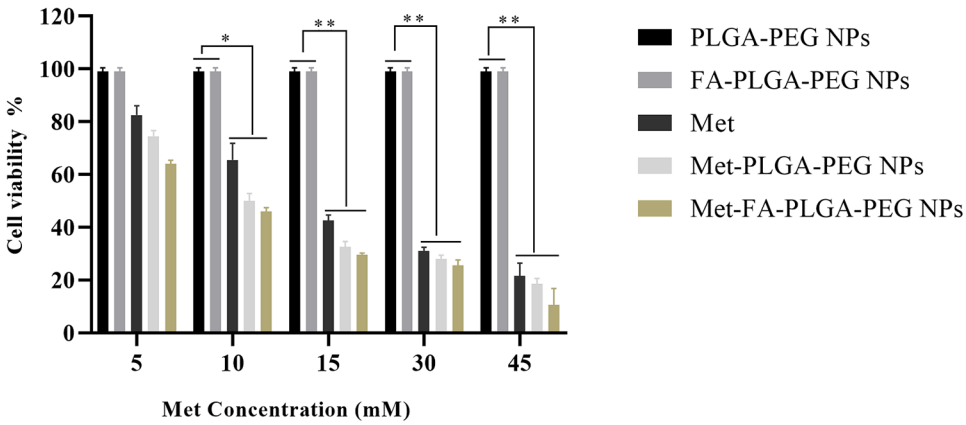

Fig. 4 MTT assay results revealed cytotoxicity of different concentrations of free Met, Met-loaded PLGA-PEG NPs, and Met loaded FA-PLGA-PEG NPs. Encapsulation of Metformin into Folate-Functionalized PLGA-PEG NPs improved its cytotoxic influence on MDA-MB-231 breast cancer cells. The results are presented as mean ± SD (n = 3). *P < 0.05, **P < 0.01, and ***P < 0.001

into the cytoplasm. Metformin accumulation (fractional uptake of dose) from FA-PLGA-PEG and PLGA-PEG NPs in the MDA-MB-231 breast cancer cells was 54.6 ± 3.4, and 39.8 ± 4.1, respectively. Results were reported as mean ± SD (n = 3). P < 0.05.

Cytotoxicity analysis

For evaluation of cytotoxicity, MTT assay was used; MDA-MB-231 cell lines were seeded in a 96-well plate and incubated for 48 h with different concentrations of Met, Met-loaded PLGA-PEG NPs, and Met-loaded FA-PLGA-PEG NPs. As shown in Fig. 4, MTT analysis demonstrated that cell survival reduces to 75% for Met treated, 40% for Met-loaded PLGA-PEG NPs, and the lowest with 21% for Met-loaded FA-PLGA-PEG NPs treated wells. As expected Met-loaded FA-PLGA-PEG NPs showed higher anti-proliferation effects against tumor cells relative to Met-loaded PLGA-PEG NPs, increased anticancer activity of the Met-loaded FA-PLGA-PEG NPs is probably related to the increased Met uptake by the cells in the case of the Met-loaded FA-PLGA-PEG NPs due to the increased internalization of the Met-loaded FA-PLGA-PEG NPs in the tumor cells.

MAD-MB-231 cell line is known as a more aggressive and invasive breast cancer line. Since the expression of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor Receptor-2 (HER2) is down-regulated, the MDA-MM-231 cell line is attributed as a triple-negative breast cancer cell line, and for this reason; the treatment option for this cell line is limited. Thus the MDA-MB-231 cell line is used to investigate new therapeutic approaches [40]. Met has been used for many years to treat type 2 diabetes due to its role in lowering blood glucose levels [41]. Recent studies have shown that Met has an anti-cancer effect and can prevent the invasion and metastasis of

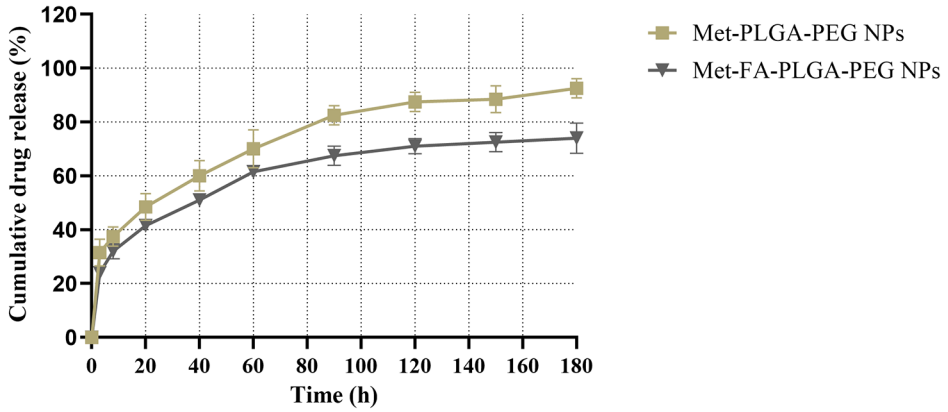

Fig. 5 The profile of in vitro release of Metformin from PLGA-PEG NPs and FA-PLGA-PEG NPs at pH = 7.4 and pH = 4.4. The results demonstrated a sustained and almost continuous release pattern from the nanoparticle with a partially burst-release at the initial stage and followed sustained during the 180 h study period

tumor cells and treat several cancers through various signaling pathways such as AMPK [42]. It has also been shown that Met exerts its anti-cancer effect on breast cancer cells by inhibiting the PI3K, Akt, and Ras-MAPK signaling pathways, which leads to reduced cell proliferation [43].

A recent study showed the anti-cancer effect of Met on MDA-MB-231. This study has demonstrated that Met-loaded PLGA-PEG NPs remarkably inhibit the growth and proliferation of MDA-MB-231 compared to Met in the free form [44].

Targeted therapies by targeting folate receptors increase treatment efficacy for malignancies and utilization of folate receptors is a new approach to cancer therapy [45]. PLGA NPs are used to deliver various types of hydrophobic and hydrophilic therapeutic agents. A recent study has shown that Folate-decorated-PLGA NPs have multifunctional properties in drug delivery and significantly have antitumor efficacy [46–48]. Another study has demonstrated that folate-decorated PEGylated copolymer for targeted drug delivery into breast cancer cells may be a promising method [49].

Results from the MTT assay were applied to calculate the concentration of drug needed to destroy 50% of tumor cells by various groups. Furthermore, after 48 h incubation of MDA-MB-231 cell lines with a 0–45 mM concentration range of metformin, the IC50 values were 13.25, 10.34, and 7.89 for Met, Met-loaded PLGA-PEG NPs and Met-loaded FA-PLGA-PEG NPs respectively (Fig. 4). Since MDA-MB-231 cells are folate receptor-positive, the higher cytotoxicity of the Met-loaded FA-PLGA-PEG NPs is perhaps related to the high affinity conjugation of FA for the folate receptor resulting in a higher cytotoxicity. This finding indicated that combination of Met-NPs enhances anti-cancer efficacy in lower doses compared to free Met [50] and decorating Met- PLGA/PEG NPs with folate remarkably can improve its anti-cancer properties and can be known as a novel and promising approaches.

Release analysis

In vitro studies of metformin release pattern demonstrated a sustained and almost continuous release pattern from the nanoparticle with a partial burst-release at the initial stage and followed sustained during the 180 h study period. This release pattern was in accordance with previous in vitro metformin release pattern studies [51].

At the end of 24 and 48 h, approximately 53% and 72.1% of Met was extricated from the PLGA-PEG NPs, respectively (Fig. 5).

The initial release is assigned to the fraction of the metformin which is adsorbed or weakly bound to the high surface area of the NPs rather than to the drugs encapsulated into the PLGA-PEG NPs. One explanation in this regard could be related to the preferred location of the drug in the particle structure. Additionally, if we extend the time of release study all of the drugs will be released from the NPs.

Cell morphological changes

After treatment with free Met, Met-loaded PLGA-PEG NPs, and Met-loaded FA-PLGA-PEG NPs the long spindle-shaped cells rounded off, and the intercellular spaces enlarged (Fig. 6).

Met-loaded FA-PLGA-PEG NPs demonstrated significant morphological changes in treated MDA-MB-231 cancer cells compared to the free Met and Met-loaded PLGA-PEG NPs.

Real-time PCR

RT-PCR technique was used to investigate Met, Met-loaded PLGA-PEG, and Met-loaded FA-PLGA-PEG NPs potential for apoptosis-inducing. For this purpose, we assessed the expression level of pro-apoptotic proteins such as Bax, Caspase7, Caspase3, p53, and anti-apoptotic proteins such

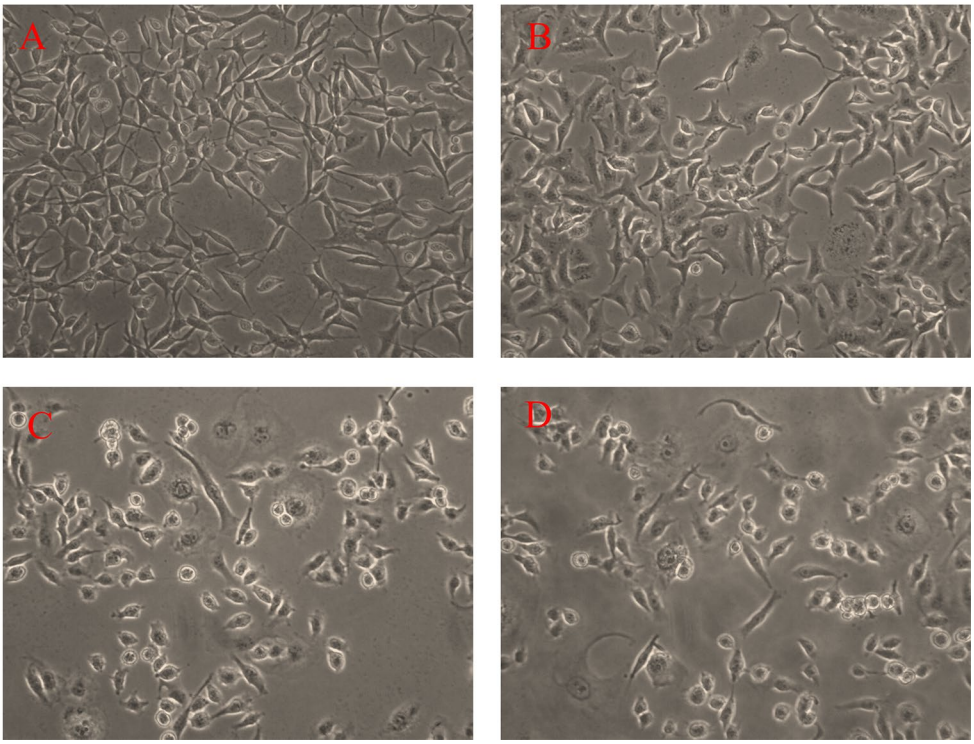

Fig. 6 Morphological changes of MDA-MB-231 cells after 48 h of incubation with Met, Met-loaded PLGA-PEG NPs, and Met-loaded FA-PLGA-PEG NPs. Control (A), free Met (B), Met loaded PLGA-PEG NPs (C), and Met loaded FA-PLGA-PEG NPs. Met-loaded FA-PLGA-PEG NPs demonstrated significant morphological changes in MDA-MB-231 cancer cells compared to the free Met and Met-loaded PLGA-PEG NPs

as hTERT and Bcl-2 [52]. The apoptosis process is performed in different ways which lead to the activation of a family of cysteine proteases, called caspase, which act as a regulator and performer of apoptosis. This process is controlled by BCL-2 family proteins which have both anti-apoptotic and pro-apoptotic effects [53]. P53 is activated when cells are exposed to malignancies which can arrest the cell cycle and induce apoptosis by interacting with the BCL-2 family [53, 54]. Therefore, manipulating the level of their gene expression can induce apoptosis and prevent cancer cell growth [14]. One of the most important features of cancer cells is the maintenance of telomere length which is performed by hTERT gene expression and telomerase. Recent studies have shown that overexpression of hTERT or the presence of hTERT mutations can be related to cancer cell migration, invasion, and metastasis [55, 56]. Also, it has been demonstrated that in 90% of breast carcinomas and in 85% of human cancers, telomerase is activated while in normal cells it is not activated. Thus, suppression of the expression of the hTERT gene and telomerase activation can be a promising approach in cancer therapy [44]. After 48 h incubation of MDA-MB-231 cell line with free, Met-loaded PLGA-PEG NPs, and Met-loaded FA-PLGA-PEG NPs, results showed that Met-loaded FA-PLGA-PEG NPs can significantly increase p53, Bax, Caspase3, and Caspase7 genes mRNA levels while reduced mRNA level of Bcl-2 and hTERT in compare of β-actin as control (Fig. 7). Also, we observed that Met-loaded FA-PLGA-PEG NPs have more effect on the expression of the studied genes than the Met-loaded PLGA-PEG NPs and free Met in MDA-MB-231 breast cancer cell lines. Javidfar et al., have shown that Met-loaded PLGA-PEG NPs significantly reduced the expression of hTERT and prevent breast cancer cells proliferation, especially MDA-MB 231 cell line which can be considered a hopeful strategy for breast cancer treatment, especially for triple-negative breast cancers. They also showed that Met-loaded PLGA-PEG NPs has more anti-cancer effect on breast cancer cells and hTERT expression than Met in the free form [44, 57].

P53 is considered a tumor suppressor for a long time, increasing p53 levels to induce apoptosis is the therapeutic strategy for cancer. For breast cancer, other studies focused on the effect of curcumin on the phosphorylation of serine residue on p53 and the activation of caspase-cascade to induce apoptosis [58, 59]. Previous studies have reported inhibition of cell proliferation and metastasis in down-regulate the expression of bcl2 and hTERT in breast cancer cell lines such as MDA-MB-231 [60].

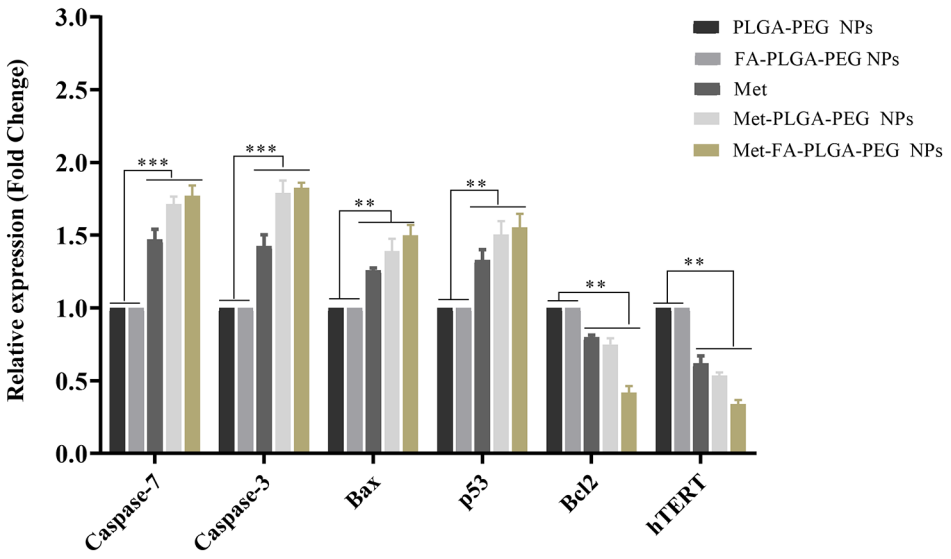

Fig. 7 Expression levels of p53, Bcl-2, Bax, Caspase7, Caspase3, and hTERT genes in MDA-MB-231 Breast cancer cells by treatment with different concentrations of pure Met, Met-loaded PLGA-PEG and, Met loaded FA-PLGA-PEG NPs. Met-loaded FA- PLGA PEG- NPs have more effect on the expression of the studied genes than the Met-PLGA PEG- NPs and free Met. *$p < 0.05$ vs. other groups was considered significant. The data are presented as mean ± SD (three independent experiments)

Conclusion

Metformin, an antidiabetic drug was subsequently discovered to show anti-cancer activity. It influenced the migration, proliferation, and survival of cancerous cells by regulating important signaling proteins. Low bioavailability and the need to use this drug in high doses to treat cancer and consequent side effects have restricted its use. This work aimed to prepare an effective carrier for Met delivery and study the influence of its folate functionalized-encapsulated form on the inhibition and proliferation of breast cancer cells. Nano-encapsulated Met was prepared by using FDA-approved copolymers in this study. NPs were acceptable in features of topography, shape, charge, size, and drug release profile. Our data proved the increase of Met anticancer effects in breast cancer cells. Our results propose that the nano-formulation of Met has improved its therapeutic indexes. Due to the intrinsic chemopreventive activity of Met, Folate functionalized NPs that are loaded with Met can be considered as a hopeful approach for developing sustainable, safe, and effective therapies for breast cancer.

Acknowledgements The research protocol was approved & Supported by Student Research Committee, Tabriz University of Medical Sciences (grant number: 65281).

Author contribution **Davoud Jafari-Gharabaghlou**: Methodology, Investigation, Data curation, Original draft preparation.

Funding The authors declare that no funds were received during the preparation of this manuscript.

Data Availability The data and materials that support the findings of this study are available from the corresponding author, upon reasonable request.

Declarations

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Mehdi Dadashpour Methodology, Reviewing and Editing.

Omid Joodi khanghah Methodology, Investigation, Original draft preparation.

Elnaz Salmani-Javan Methodology, Investigation.

Nosratollah Zarghami Supervision, Conceptualization, Writing- Reviewing and Editing.

Competing Interests No potential competing interest was reported by the authors.

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